Campus Event Management Platform

CS 202C course project ▪ IIT Dharwad

Test Plan & Report

Testing strategy, test cases and results

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Part of the project documentation set: Project Report • Presentation Slides • User Manual • Installation & Operations Guide • Developer Guide • Test Plan & Report. Source: `docs/` and `report/` in the project repository.

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1 Summary

Automated test cases	**45**
Passed	**45**
Failed / errors / skipped	0 / 0 / 0
Run time	about 7 seconds
Line coverage of `app/`	76% (`reports.py` 100%)

Table 1: Result of the latest full run (10 November 2026).

All 45 automated tests pass. They cover authentication and roles, browsing and searching, the registration rules, admin management with validation and the venue clash check, participant CSV export, and the exact numbers produced by the three reports. Untested code is mainly the demo-data command (`seed.py`) and some admin error paths, listed in Section 6.

2 Test plan

2.1 Objectives

1. Confirm that every feature in the requirements works as specified.
2. Confirm that every *refusal* works: invalid input, full or started events, duplicate registrations, venue clashes and access by the wrong role are rejected and change nothing.
3. Protect the profiling optimisation: the reports must return exactly the same numbers after the query rewrite and with or without indexes.
4. Make tests fast and repeatable so they can be run before every commit.

2.2 Scope

In scope All routes in `auth`, `events` and `admin`; the helper functions `find_clash`, `percent` and the three report functions; database constraints that the code relies on (unique registrations, cascade delete).

Out of scope for automation Visual layout and CSS, browser `confirm()` dialogs, the Bash scripts, the `seed` command, and performance (measured separately with `cProfile`, see the project report). These are covered by the manual checklist in Section 7.

2.3 Approach

Level. Mostly *functional* tests through Flask's test client, which sends real HTTP requests to the application without starting a server, so each test exercises routing, login, validation, SQL and templates together. A few *unit* tests call plain functions (`find_clash`, `percent`, the report functions) directly.

Isolation. The `app` fixture creates a new application for *every* test with its database in pytest's per-test temporary folder, and runs `sql/schema.sql` on it. Tests can never see each other's data or touch `instance/campus.db`.

Assertions on outcomes. Tests check both what the user sees (status code, redirect target, flash message text) and what is stored (a direct SQL query afterwards). A refused action must leave the row count unchanged.

Exact expected values. Report tests compare against hand-calculated numbers. They check *what* is computed, not *how*, so they passed unchanged before and after the optimisation.

2.4 Test environment

Item	Value
Framework	pytest 8 with Flask's test client; configuration in `pytest.ini` (`testpaths = tests`, `pythonpath = .`)
Command	`./scripts/test.sh` (runs `python -m pytest -v`)
Database	a fresh SQLite file per test in `tmp_path`
Environments	Windows 11 + Git Bash + Python 3.14 (development machine); Linux + Python 3.13 (run recorded in Section 5)

Table 2: Test environment.

2.5 Fixtures and test data

Fixture	Provides
`app`	application with `TESTING=True` and an empty temporary database
`client`	test client (fake browser)
`auth`	`register()` / `login()` helpers; default student `student@test.com` / `secret123`
`admin_user`	admin `admin@test.com` / `adminpass`, inserted with SQL
`sample_events`	venue "Seminar Hall A" (capacity 100) and four events, returned as `{title: id}`
`student_client`	client logged in as the default student
`admin_client`	client logged in as the admin
`report_data`	(in `test_reports.py`) adds an empty venue and 6 registrations with known counts

Table 3: Fixtures (`tests/conftest.py`).

Event	Category	Starts	Capacity	Why it exists
Python Workshop	Workshop	in 2 days	30	"Python" in the title
Football Match	Sports	in 3 days	50	does not match "python"
Hackathon	Technical	in 4 days	40	"Python" only in the description
Old Python Talk	Seminar	3 days ago	40	past: must never be listed or bookable

Table 4: The `sample_events` data set. Each event lasts 0.1 day (2.4 hours).

3 Test cases

Each case lists its ID, the pytest function (in the file named in the subsection title) and the scenario with the expected result. All cases passed in the run recorded in Section 5.

3.1 Authentication and roles (`test_auth.py`, 7 tests)

ID	Test function	Scenario and expected result
`AUTH-01`	`test_register_creates_student`	Register a new account → 302 redirect to `/auth/login`; the row exists with role `student`; the stored hash is not the plain password.
`AUTH-02`	`test_duplicate_email_rejected`	Register twice with the same email → "already exists"; still exactly one user with that email.

ID	Test function	Scenario and expected result
AUTH-03	test_login_with_correct_password_works	Log in with the right password → 302; the events page now shows *Logout*.
AUTH-04	test_login_with_wrong_password_fails	Wrong password → 200 on the login page with "Invalid email or password"; not logged in.
AUTH-05	test_student_gets_403_on_admin_route	Student opens /admin/ → 403.
AUTH-06	test_admin_can_open_admin_route	Admin opens /admin/ → 200 with "Admin dashboard".
AUTH-07	test_logged_out_user_is_sent_to_login_from_admin_route	Visitor opens /admin/ → 302 to the login page.

3.2 Browsing and search (test_events.py, 5 tests)

ID	Test function	Scenario and expected result
EVT-01	test_event_list_shows_only_upcoming_events	/events/ lists Python Workshop and Football Match but not the past Old Python Talk.
EVT-02	test_keyword_search_returns_only_matching_events	?q=python matches the title (case-insensitive) and the description (Hackathon); excludes Football Match and the past event.
EVT-03	test_category_filter	?category=Sports shows Football Match only.
EVT-04	test_seats_left_counts_registrations	One registration on a 30-seat event → detail page shows "29 of 30 left".
EVT-05	test_missing_event_returns_404	/events/9999 → 404.

3.3 Registrations (test_registrations.py, 8 tests)

ID	Test function	Scenario and expected result
REG-01	test_register_for_event	Student registers → "You are registered", "29 of 30 left", one row stored.
REG-02	test_double_registration_rejected	Second registration → "already registered"; still one row.
REG-03	test_full_event_rejected	Capacity 1 already taken by another student → "full"; count stays 1.
REG-04	test_started_event_rejected	Register for the past event → "already started"; no row.
REG-05	test_cancel_registration	Register then cancel → "was cancelled"; no row.
REG-06	test_my_registrations_lists_only_my_events	My registrations shows the registered event and not others.
REG-07	test_cancel_from_my_registrations_returns_there	Cancel with return_to=my → redirect to /my-registrations.
REG-08	test_must_be_logged_in_to_register	Visitor POSTs to register → redirect to login.

3.4 Administration (test_admin.py, 18 tests)

ID	Test function	Scenario and expected result
ADM-01	test_find_clash	Unit test of the overlap rule against the workshop: identical times, overlap at the start and fully inside all clash; ending exactly at its start, a different venue, and ignoring the event itself do not.
ADM-02	test_create_event	Admin creates an event → "created"; 2030-05-01T10:00 is stored as 2030-05-01 10:00.

ID	Test function	Scenario and expected result
ADM-03	test_venue_clash_rejected	New event at the workshop's venue and times → "already booked"; event count unchanged.
ADM-04	test_edit_does_not_clash_with_itself	Edit only the title of an event → "updated"; new title stored.
ADM-05	test_capacity_over_venue_capacity_rejected	Capacity 500 at a 100-seat venue → "cannot be more than the capacity"; still 4 events.
ADM-06	test_end_before_start_rejected	End before start → "must end after it starts".
ADM-07	test_delete_event_removes_its_registrations	Delete an event with a registration → event gone and its registrations removed (ON DELETE CASCADE).
ADM-08	test_create_venue	Create venue "New Hall" capacity 75 → listed and stored.
ADM-09	test_venue_with_events_cannot_be_deleted	Delete a venue in use → "Cannot delete"; venue still exists.
ADM-10a-g	test_student_cannot_open_admin_pages (parametrised, 7 cases)	A student GETs each of /admin/venues, /admin/venues/new, /admin/events, /admin/events/new, /admin/events/1/edit, /admin/events/1/participants, /admin/events/1/participants.csv → 403 every time.
ADM-11	test_student_cannot_delete_event	Student POSTs delete → 403; event still exists.
ADM-12	test_csv_export_returns_the_right_rows	Three students, two on the workshop → CSV is text/csv, an attachment, and contains exactly the header plus Asha and Ravi in registration order (not Meera).

3.5 Reports (test_reports.py, 7 tests)

The report_data fixture registers 3 students for Python Workshop, 2 for Old Python Talk, 1 for Football Match and none for Hackathon, and adds an empty venue "Empty Hall".

ID	Test function	Scenario and expected result
REP-01	test_percent	percent(3,30) = 10.0, percent(1,3) = 33.3, percent(5,0) = 0.0 (no division by zero).
REP-02	test_event_stats	Exactly: Workshop 3/30 = 10.0%, Old Talk 2/40 = 5.0%, Football 1/50 = 2.0%, Hackathon 0/40 = 0.0%, in that order.
REP-03	test_popular_categories	Exactly: Workshop (1 event, 3), Seminar (1, 2), Sports (1, 1), Technical (1, 0).
REP-04	test_venue_utilisation	Exactly: Seminar Hall A 4 events, 6 registrations, 160 seats, 3.8%; Empty Hall 0, 0, 0, 0.0% (listed even with no events).
REP-05	test_indexes_do_not_change_report_numbers	All three reports are identical before and after add_indexes(), and the three idx_ indexes exist.
REP-06	test_reports_page_shows_numbers	Admin opens /admin/reports → 200 with "Python Workshop", "10.0%" and "Empty Hall".
REP-07	test_student_cannot_open_reports	Student → 403.

4 Requirements traceability

Requirement	Covered by
Students can create an account; emails are unique	AUTH-01, AUTH-02
Passwords are stored hashed	AUTH-01

Requirement	Covered by
Login with correct / wrong credentials	AUTH-03, AUTH-04
Admin pages require an admin (403 for students, login for visitors)	AUTH-05–07, ADM-10a–g, ADM-11, REP-07
Browse upcoming events only	EVT-01
Search by keyword and filter by category	EVT-02, EVT-03
Seats left are correct	EVT-04, REG-01
Register for an event	REG-01
Refuse duplicate / full / started registrations; require login	REG-02, REG-03, REG-04, REG-08
Cancel a registration; My registrations	REG-05, REG-06, REG-07
Admin creates and edits events	ADM-02, ADM-04
Event validation (times, capacity vs. venue)	ADM-05, ADM-06
No venue time clashes	ADM-01, ADM-03, ADM-04
Deleting an event removes its registrations	ADM-07
Venue management; venue in use cannot be deleted	ADM-08, ADM-09
Participant list exported as CSV	ADM-12
Reports: per-event fill rate, categories, venue utilisation	REP-01–04, REP-06
Optimisation (rewrite + indexes) does not change any report number	REP-02–05
Missing pages return 404	EVT-05

5 Test results

5.1 Latest run

Run on 10 November 2026 on Linux with Python 3.13, using `python -m pytest -v` (the same command `./scripts/test.sh` runs). Output abridged: lines for repeated tests in the same file are summarised.

```
collected 45 items

tests/test_admin.py::test_find_clash PASSED                                [  2%]
tests/test_admin.py::test_create_event PASSED                              [  4%]
tests/test_admin.py::test_venue_clash_rejected PASSED                      [  6%]
tests/test_admin.py::test_edit_does_not_clash_with_itself PASSED           [  8%]
tests/test_admin.py::test_capacity_over_venue_capacity_rejected PASSED     [ 11%]
tests/test_admin.py::test_end_before_start_rejected PASSED                 [ 13%]
tests/test_admin.py::test_delete_event_removes_its_registrations PASSED    [ 15%]
tests/test_admin.py::test_create_venue PASSED                              [ 17%]
tests/test_admin.py::test_venue_with_events_cannot_be_deleted PASSED       [ 20%]
tests/test_admin.py::test_student_cannot_open_admin_pages[...] PASSED x7   [ 35%]
tests/test_admin.py::test_student_cannot_delete_event PASSED               [ 37%]
tests/test_admin.py::test_csv_export_returns_the_right_rows PASSED         [ 40%]
tests/test_auth.py::... 7 PASSED                                           [ 55%]
tests/test_events.py::... 5 PASSED                                         [ 66%]
tests/test_registrations.py::... 8 PASSED                                  [ 84%]
tests/test_reports.py::... 7 PASSED                                        [100%]

============================== 45 passed in 6.53s ==============================
```

File	Tests	Passed	Result
`test_auth.py`	7	7	**PASS**
`test_events.py`	5	5	**PASS**
`test_registrations.py`	8	8	**PASS**
`test_admin.py`	18	18	**PASS**
`test_reports.py`	7	7	**PASS**
Total	**45**	**45**	**PASS**

Table 5: Results per file.

5.2 Run on the development machine

The suite was also run throughout development in Git Bash on Windows. Figure 1 shows a run from the profiling phase, when the suite had 44 tests (REP-05 was added with the indexes).

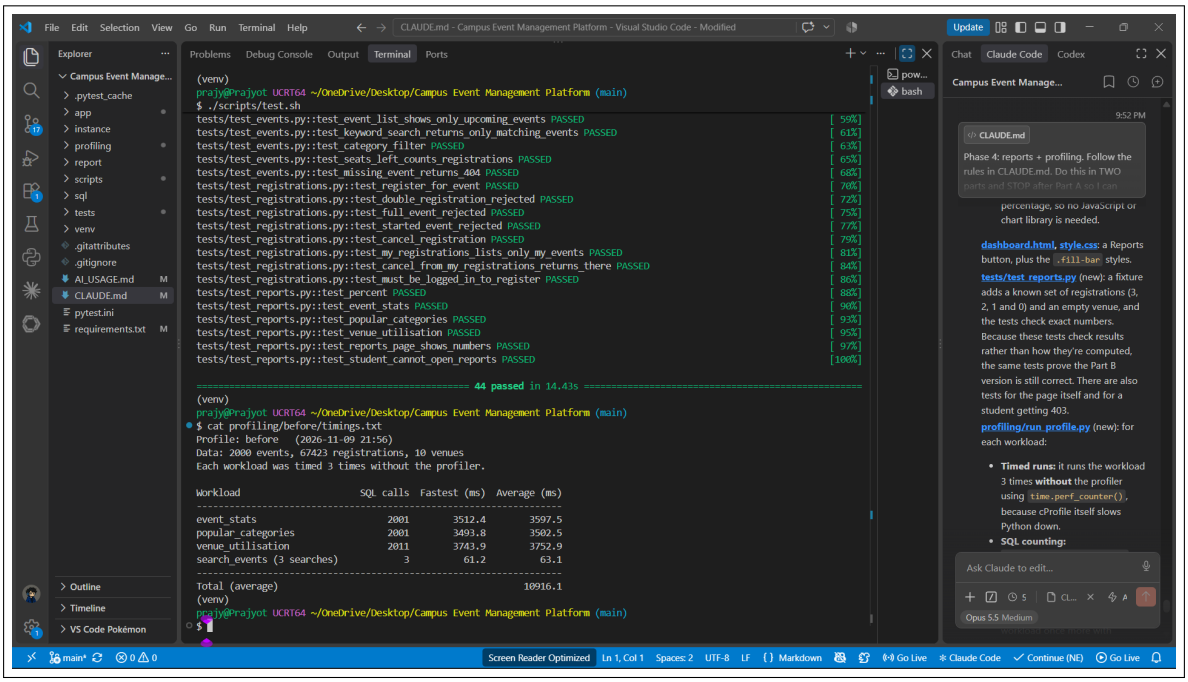


Figure 1: `./scripts/test.sh` in the VS Code terminal: 44 passed, followed by the baseline profiling timings.

5.3 History

Phase	Tests	Added
2. Auth and browsing	12	fixtures, `test_auth.py`, `test_events.py`
3. Registrations and admin	38	`test_registrations.py`, `test_admin.py`
4a. Naive reports	44	`test_reports.py` (6 tests)
4b. Optimisation	45	REP-05: identical numbers with and without indexes

Table 6: Growth of the test suite. Every phase ended with all tests passing.

Defect found by testing. The first version of ADM-01 used a window starting in the year 2000 and expected no clash. It failed because that window genuinely overlapped the past sample event at the same venue: the application was right and the test was wrong. The test now uses

a one-hour window ending exactly when the workshop starts, which also checks that touching intervals are allowed.

6 Code coverage

Measured with `pytest-cov` (`python -m pytest -cov=app`); `pytest-cov` is not a project dependency and is only needed to repeat this measurement.

Module	Statements	Missed	Coverage
`app/__init__.py`	27	3	89%
`app/admin.py`	203	43	79%
`app/auth.py`	81	9	89%
`app/db.py`	52	13	75%
`app/events.py`	98	11	89%
`app/reports.py`	24	0	100%
`app/seed.py`	67	51	24%
Total	**552**	**130**	**76%**

Table 7: Line coverage by module.

The main gaps are:

- `seed.py`: the demo-data command is exercised by `seed.sh` and `profile.sh`, not by pytest.
- `db.py`: the CLI commands `init-db`, `add-indexes` and `create-admin` (the functions behind them are tested through fixtures).
- `admin.py`: some validation messages (missing title, category or venue, invalid times, lowering capacity below registrations, venue edit rules), venue edit and the HTML participants page.
- `events.py`: cancel after start, cancel when not registered, admin trying to register, and the venue/date filters through the route.

These paths are short, follow the same pattern as tested ones, and are covered by the manual checklist below; they are candidates for future automated tests.

7 Manual acceptance checklist

Items that need a real browser or a person's judgement. Run against `./scripts/setup.sh`, `./scripts/seed.sh` and `./scripts/run.sh`; tick each item when checked.

ID	Step and expected result	OK
MAN-01	Navigation bar shows the right links for visitor, student and admin.	☐
MAN-02	Search by venue and by date range returns only matching events.	☐
MAN-03	Search for `' OR 1=1 -` returns no events and no error (SQL injection attempt).	☐
MAN-04	An event description containing `<b>bold</b>` is shown as text, not bold (HTML escaping).	☐
MAN-05	Delete buttons ask for confirmation; choosing *Cancel* deletes nothing.	☐
MAN-06	Edit an event and lower its capacity below the number registered: refused with a clear message.	☐
MAN-07	A downloaded participants CSV opens correctly in Excel/LibreOffice, including names with commas.	☐
MAN-08	Reports page fill-rate bars match the percentages.	☐

ID	Step and expected result	OK
MAN-09	At phone width (DevTools, 375 px) cards stack and wide tables scroll sideways.	☐
MAN-10	Opening `/admin/` in a private window redirects to the login page.	☐
MAN-11	`./scripts/seed.sh abc` prints an error and changes nothing.	☐
MAN-12	`./scripts/profile.sh after` completes and writes the four result files.	☐

8 Known limitations and future tests

- **Concurrency.** Two simultaneous registrations for the last seat are not tested (the check and insert are separate statements). A future test could run two registrations in threads after the registration is made atomic.
- **Time-dependent data.** Fixtures place events relative to *now*, so tests are stable on any date, but a run crossing midnight in the exact second an event starts is theoretically possible.
- **Browser tests.** No end-to-end browser automation (e.g. Playwright); layout is checked manually.
- **Performance** is not asserted in tests; it is measured with `profile.sh` and recorded in `profiling/RESULTS.md`.